

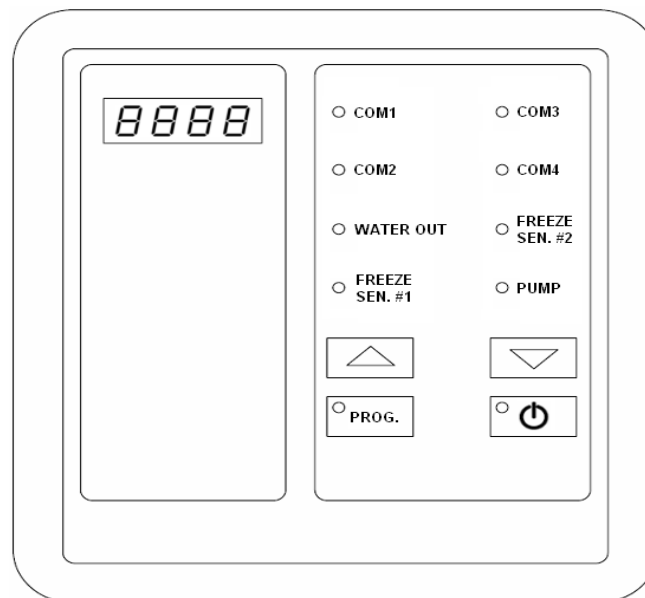
1. INTRODUCTION

The Intronics Chiller Control System (ICCS) is a micro-processor based controller which provides all control, protection and self diagnostic features for a chiller system with simplified wiring and an easy to use control panel.

The air conditioner can use either hermetic or semi-hermetic type compressor(s). The condensing unit can be up to 2 refrigerating circuits with 2 compressors per circuit.

ICCS consists of

- ❑ Main control board
- ❑ Control panel unit (up to 2 units) with standard 6 meters length cable.
- ❑ 3 temperature sensors - water out, water freeze 1 & water freeze 2 with standard 7 meter length.



The main board and control panel are connected by a 4 core cable with RS485 connectors. Distance between the main board and control panel can be a maximum of 1000m. Since two RS485 connectors are provided on the main board, a total of 2 Control panel units can be installed.

There are a total of 12 inputs and outputs (3 Analog Inputs & 9 Digital Outputs).

Inputs

- ❑ Water temperature Sensor
- ❑ Freeze 1 sensor for refrigerating circuit 1
- ❑ Freeze 2 sensor for refrigerating circuit 2

Outputs

- ❑ Pump chiller relay
- ❑ Pump supply relay
- ❑ Alarm relay
- ❑ Fan motor 1 relay (Circuit #1)
- ❑ Fan motor 2 relay (Circuit #2)
- ❑ Compressor contactor relay for Circuit 1A (C_{1A}) (Circuit #1)
- ❑ Compressor contactor relay for Circuit 1B (C_{1B}) (Circuit #1)
- ❑ Compressor contactor relay for Circuit 2A (C_{2A}) (Circuit #2)
- ❑ Compressor contactor relay for Circuit 2B (C_{2B}) (Circuit #2)

Fig.1 is the block diagram of the system.

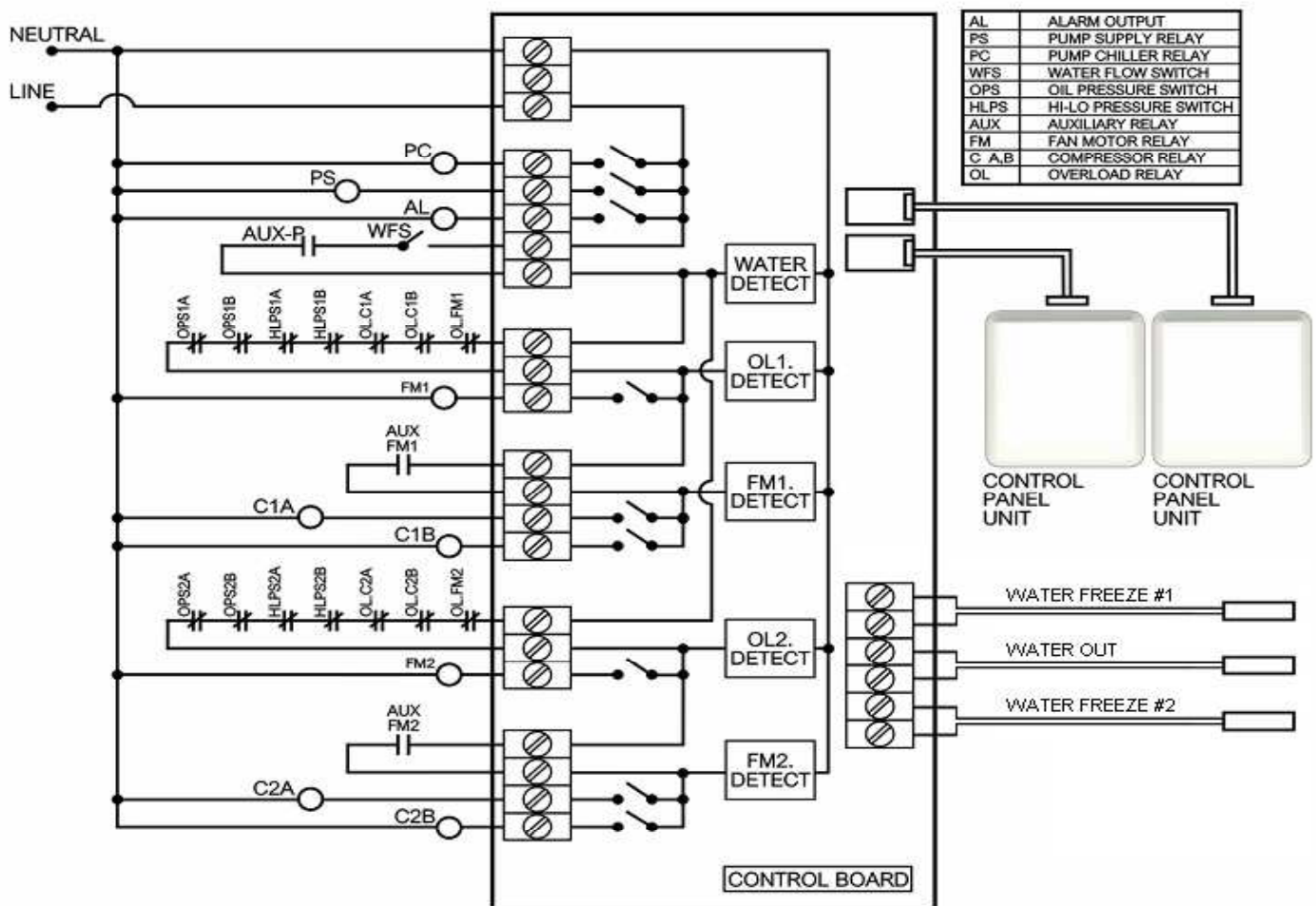


Fig. 1 System block diagram

2. FUNCTION

2.1 Water Supply

Press  button to turn the system on and off. Turning On will turn on the Water pump chiller relay and Water pump supply relay. After a delay time, according to a value set in Program **PG: 02**, the system will start to control the operation of the Fan motors and Compressors.

2.2 Temperature control

Temperature control is achieved by control of the compressors to regulate the temperature of Water out sensor (Tw) when compared to a set temperature (Ts) according to a value set in Program **PG: 01**. The system will control the operating sequence and the number of compressors to regulate the temperature as follows:

- ❑ The number of compressors involved is based on the following stages
 - Off Stage : All compressor units are turned off
 - Stage 1: Only one unit will be on.
 - Stage 2: Two units will be on (type 2 or 4 Compressor)
 - Stage 3: Three units will be on (type 4 Compressor)
 - Stage 4: Four units will be on (type 4 Compressor)
- ❑ The sequences of the working compressors are based on the stage of each Model (1, 2 or 4 Compressors / 1 or 2 Circuits). Models can be selected by dipswitches, SW1.1 and SW1.2, on the board.
- ❑ For the four Compressor model, the compressor sequence will depend on the values set in Program **PG: 03**.
- ❑ The system will swap the compressor sequence once all the compressor units have been turned on (based on the number of units set from Model Selection Option), or all the compressor units are turned off.

4-Comp. / 2-Circuit PG:03 = "1"		Compressor (Normal)				Compressor (Swap)			
		C1A	C1B	C2A	C2B	C1A	C1B	C2A	C2B
Turn On Sequence		1	2	3	4	4	3	2	1
Stage 4	Tw-Ts $\geq$ 2.0C	ON	ON	ON	ON	ON	ON	ON	ON
Stage 3	Tw-Ts = 1.0C	ON	ON	ON	OFF	OFF	ON	ON	ON
Stage 2	Tw-Ts = 0.0C	ON	ON	OFF	OFF	OFF	OFF	ON	ON
Stage 1	Tw-Ts = -1.0C	ON	OFF	OFF	OFF	OFF	OFF	OFF	ON
Off Stage	Tw-Ts $\leq$ -2.0C	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF

4-Comp. / 2-Circuit PG:03 = "0"		Compressor (Normal)				Compressor (Swap)			
		C1A	C1B	C2A	C2B	C1A	C1B	C2A	C2B
Turn On Sequence		1	3	2	4	4	2	3	1
Stage 4	Tw-Ts $\geq$ 2.0C	ON	ON	ON	ON	ON	ON	ON	ON
Stage 3	Tw-Ts = 1.0C	ON	ON	ON	OFF	OFF	ON	ON	ON
Stage 2	Tw-Ts = 0.0C	ON	OFF	ON	OFF	OFF	ON	OFF	ON
Stage 1	Tw-Ts = -1.0C	ON	OFF	OFF	OFF	OFF	OFF	OFF	ON
Off Stage	Tw-Ts $\leq$ -2.0C	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF

2-Comp. / 2-Circuit		Compressor (Normal)				Compressor (Swap)			
		C1A	C1B	C2A	C2B	C1A	C1B	C2A	C2B
Turn On Sequence		1	-	2	-	2	-	1	-
Stage 2	Tw-Ts $\geq$ 1.0C	ON	N.A.	ON	N.A.	ON	N.A.	ON	N.A.
Stage 1	Tw-Ts = 0.0C	ON	N.A.	OFF	N.A.	OFF	N.A.	ON	N.A.
Off Stage	Tw-Ts $\leq$ -1.0C	OFF	N.A.	OFF	N.A.	OFF	N.A.	OFF	N.A.

2-Comp. / 1-Circuit		Compressor (Normal)				Compressor (Swap)			
		C1A	C1B	C2A	C2B	C1A	C1B	C2A	C2B
Turn On Sequence		1	2	-	-	2	1	-	-
Stage 2	Tw-Ts $\geq$ 1.0C	ON	ON	N.A.	N.A.	ON	ON	N.A.	N.A.
Stage 1	Tw-Ts = 0.0C	ON	OFF	N.A.	N.A.	OFF	ON	N.A.	N.A.
Off Stage	Tw-Ts $\leq$ -1.0C	OFF	OFF	N.A.	N.A.	OFF	OFF	N.A.	N.A.

1-Comp. / 1-Circuit		Compressor (Normal)				Compressor (Swap)			
		C1A	C1B	C2A	C2B	C1A	C1B	C2A	C2B
Stage 1	Tw-Ts $\geq$ 1.0C	ON	N.A.	N.A.	N.A.	ON	N.A.	N.A.	N.A.
Off Stage	Tw-Ts $\leq$ 0.0C	OFF	N.A.	N.A.	N.A.	OFF	N.A.	N.A.	N.A.

Note :

- Tw : Water out temperature
- Ts : Setting temperature (PG: 01)
- C1A, C1B is same refrigerant circuit (Circuit #1)
- C2A, C2B is same refrigerant circuit (Circuit #2)

Example:

- Turn Off sequence will depend on the Turn On sequence.

Turn On sequence: $C1A \rightarrow C2A \rightarrow C1B \rightarrow C2B$

$C1A \leftarrow C2A \leftarrow C1B \leftarrow C2B$: Turn Off sequence

Alternative sequence for the above would be the following:

Turn On sequence: $C2B \rightarrow C1B \rightarrow C2A \rightarrow C1A$

$C2B \leftarrow C1B \leftarrow C2A \leftarrow C1A$: Turn Off sequence

- Another example would be the following.

Turn On sequence: $C1A \rightarrow C1B \rightarrow C2A \rightarrow C2B$

$C1A \leftarrow C1B \leftarrow C2A \leftarrow C2B$: Turn Off sequence

Alternative sequence for the above would be the following:

Turn On sequence : $C2B \rightarrow C2A \rightarrow C1B \rightarrow C1A$

$C2B \leftarrow C2A \leftarrow C1B \leftarrow C1A$: Turn Off sequence

- ❑ There will be a delay time for each compressor to be turned on as set in the program **PG: 06** (Compressor turn-on separation delay time)

2.3 Fan motor

- ❑ The system will turn on the Fan motor of circuit #1 when the Compressor of circuit #1 (C1A or C1B) is turned on.
- ❑ The system will turn on the Fan motor of circuit #2 when the Compressor of circuit #2 (C2A or C2B) is turned on.

2.4 Water supply error

If there is a problem with the water supply in the Chiller unit, which can be sensed by

- ❑ The magnetic contactor of the water pump not energizing, or
- ❑ The water flow switch does not operate;

the system will turn off the Water pump, Fan motors and Compressors of all refrigerating circuits.

The Control panel will show **Er. 01** flashing.

- ❑ Whenever the Water supply error occurs, the control needs to be reset by turning off and on. The control will then resume normal operation.

2.5 Overload error

- ❑ If the system detects a problem in the Overload detector circuit of Cooling system circuit #1 (Overload protection of Fan motor or compressors, or High/Low pressure switch, or Oil pressure switch of Cooling system circuit #1 is on), the system will turn off the Fan motor and Compressors of Circuit #1 and displays **Er. 02** flashing at the control panel.
- ❑ In the case of an Overload error in the Cooling system circuit #2, the control panel will show **Er. 04** flashing.
- ❑ Whenever the Overload error occurs, the control needs to be reset by turning off and on. The control will then resume normal operation.

2.6 Fan motor error (AUX circuit)

- ❑ If the system turns on the Fan motor of circuit #1 and detects an open contact in the magnetic contactor of Fan motor circuit #1, the system will turn off the Fan motor and Compressors of circuit #1 and display **Er. 03** flashing at the control panel.
- ❑ In the case of Fan motor error in circuit #2, the control panel will show **Er. 05** flashing.
- ❑ Whenever the Fan motor error occurs, the control needs to be reset by turning off and on. The control will then resume normal operation.

2.7 Water freeze protection

- ❑ While the system is on, IF the temperature of the water freeze sensor #1 or the water freeze sensor #2 or the water out sensor is less than or equal to the setpoint temperature in Program **PG: 05**, the system will turn off the fan motor and compressor to prevent ice from forming. On the control panel, the LED light next to Water freeze sensor #1 or Water freeze sensor #2 or Water out sensor will flash depending on which sensor detected the freeze temperature.
- ❑ The system will resume its normal operation once the temperature of Water freeze sensor #1 and Water freeze sensor #2 and Water out sensor is more than or equal to 6.5° C (44° F).
- ❑ If the freeze condition occurs more than 3 times and each time is less than 60 minutes, the system will turn off all the functions and display **Er. 07** flashing at the control panel. The control needs to be reset or the Power supply needs to be turned off and on. The control will then resume normal operation.
- ❑ If the freeze condition is not detected for longer than 60 minutes or the system is turned off, the counter will be reset.

2.8 Water out temperature sensor error

If the value detected by the Water out temperature sensor is outside its normal range,

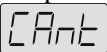
- ❑ The Control will turn off the water pump, compressor and fan.
- ❑ The Control panel will show **Er. 08** flashing.
- ❑ Whenever the Water out temperature sensor error occurs, the control needs to be reset by turning off and on. The control will then resume normal operation.

2.9 Water freeze temperature sensor error

If the value detected by the water freeze sensor is outside its normal range,

- ❑ The Control will turn off the water pump, compressor and fan.
- ❑ The Control panel will show **Er. 09** or **Er. 10** flashing.
- ❑ Whenever the freeze temperature sensor error occurs, the control needs to be reset by turning off and on. The control will then resume normal operation.

2.10 Communication error

If there is an error in the communication between the Main control board and the Control panel unit, for example the communication line is disconnected, the Control panel will show  flashing.

2.11 Alarm output

The Alarm relay will be energized if there is any system error.

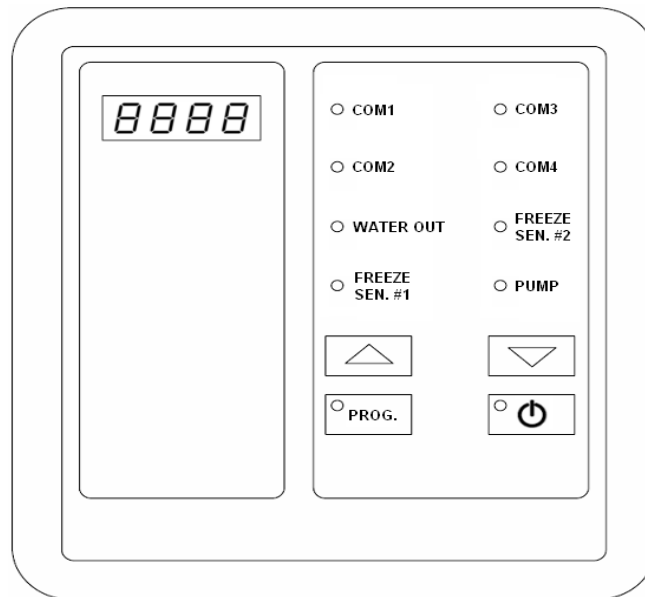
2.12 Compressor delay protection

There is a 3 minute time delay for the compressor to restart.

2.13 Program Error

If there is any program set point value in system memory that is not valid, the system will display **Er. 06** flashing along with the invalid value, until the correct program value is set.

3. USER INTERFACE



3.1 Four digit LED display

- ❑ Shows the temperature reading of Water out sensor, Water freeze sensor #1 or freeze sensor #2. Shows **Lo** when the temperature reading is lower than -8.0°C (18°F) and shows **Hi** when the temperature reading is higher than 47.0°C (117°F). Temperature increment is 0.5°C or 1°F .
- ❑ Shows various setpoint values.
- ❑ Shows various error codes, ex. Water freeze error.

3.2 LED display status

- ❑ Compressor 1,2,3,4 On/Off status
 - 4-Comp / 2-Circuit Model : Comp 1,2,3,4 = C1A,C1B,C2A,C2B
 - 2-Comp / 2-Circuit Model : Comp 1,2 = C1A,C2A
 - 2-Comp / 1-Circuit Model : Comp 1,2 = C1A,C1B
 - 1-Comp / 1-Circuit Model : Comp 1 = C1A

- ❑ Water out temperature display status
- ❑ Water freeze sensor #1 temperature display status
- ❑ Water freeze sensor #2 temperature display status
- ❑ Pump On/Off status
- ❑ Program setting status
- ❑ Power On/Off status

3.3 Pushbuttons

- ❑ Press Power button to turn the system on/off.
- ❑ Press Up or Down button while the system is on to select each sensor to show its temperature values on the display.
- ❑ Press PROG button while the system is on or off to enter programmable mode as follows:
 - Press and hold PROG for 5 second till the LED at PROG button is flashing.
 - Press POWER while still holding the PROG button, the display will show the last programmable value. Let go off both buttons, the PROG LED will switch from flashing to steady, this means that the system is in programmable mode.
 - Press PROG button to choose each program in sequence. There are 6 programs from **PG: 01** to **PG: 06**.
 - After selecting a program, press UP or DOWN button to set the current setting values.
 - Press PROG button to go to the next program sequence.
 - You can press POWER button at any time to leave the programmable mode.
 - In case there is no button press for more than 30 seconds, the system will leave this mode automatically.

4. PROGRAMMING

4.1 Program "P1"

PG: 01

- ☐ To set the Water out temperature
- ☐ The value can be set from 4.5 °C to 24.0 °C at 0.5 °C increments
(40° F to 75° F at 1 °F increments)
- ☐ Factory setting = 7.0 °C (45 °F)

4.2 Program "P2"

PG: 02

- ☐ To set the time delay for the Water supply. Temperature control will activate after the delay.
- ☐ The delay can be set from 1 min to 30 min at 1 min increments.
- ☐ Factory setting = 1 minute

4.3 Program "P3"

PG: 03

- ☐ To set up the working sequence of the 4 compressors model.
 - "0" = C1A → C2A → C1B → C2B
 - "1" = C1A → C1B → C2A → C2B
- ☐ Factory setting = "0"
- ☐ This program has no effect on either 1 or 2 compressor models.

4.4 Program "P4"

PG: 04

- ☐ To set the temperature scale to either °C or °F
- ☐ Factory setting = °C

4.5 Program “P5”

PG: 05

- ☐ To set the temperature value for Freeze detection (Water freeze temperature)
- ☐ The value can be set from -8.0 °C to 4.0 °C at 0.5 °C increments (18 °F to 39 °F at 1 °F increments)
- ☐ Factory setting = -2.0 °C (28 °F)

4.6 Program “P6”

PG: 06

- ☐ To set the Compressor turn-on separation delay time
- ☐ The delay can be set between 1 min to 10 min at 1 min increments
- ☐ Factory setting = 1 min.

5. LED ON MAIN BOARD

5.1 LED 1-4

To show the status of compressor C1A, C1B, C2A, C2B respectively.

5.2 LED 5

To show the working status of the system. Continuous flashing indicates that the system is working properly. There is a system malfunction if the LED stops flashing.

5.3 LED 6

LED on indicates that the transformer is providing the correct voltage. If the LED is off, there is no voltage from the transformer to the board.

6. OPTION

6.1 Test mode

- ❑ If dip switch SW1.4 is set to ON, the control board will operate with
 - No compressor time delay, both in changing stages and compressor delay protection
 - No waiting time after water pump is turned on as set in Program **PG: 02**

6.2 Auto-restart / Force power on

- ❑ Auto-Restart: In case of power outage, the system will maintain its previous setting.
- ❑ Force Power On: If dip switch SW1.3 is set to ON, the system will be forced to turn on regardless of system status prior to power outage.

6.3 Model Selection

By setting dip switch SW1.1 and SW1.2 as follows:

SW1.1	SW1.2	MODEL
OFF	OFF	4-Compressor / 2-Circuit
ON	OFF	2-Compressor / 2-Circuit
OFF	ON	2-Compressor / 1-Circuit
ON	ON	1-Compressor / 1-Circuit

- ❑ In case of 1-Circuit model, the system will not detect Overload error and Fan motor error of circuit #2.
- ❑ If the compressor has been turned off for less than 3 minutes, that unit will not be turned on. The unit that has been turned off for 3 minutes will be returned to operation first (Compressor protection delay time).